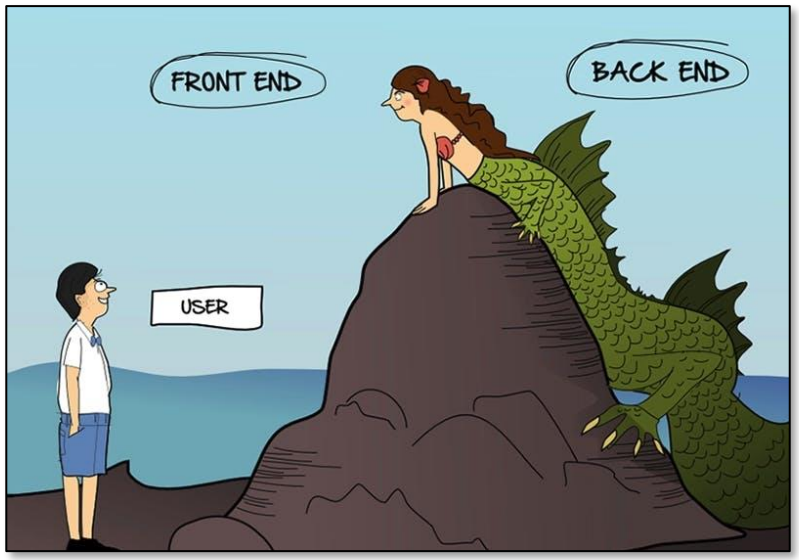


인터넷 프로그래밍 및 응용 교과목 소개



서강대학교
Sogang University

- 강의 구성

I. Internet & Web Architecture

1) TCP-IP 프로토콜 및 인터넷
2) HTML, HTTP, CSS

II. LAMP Stack

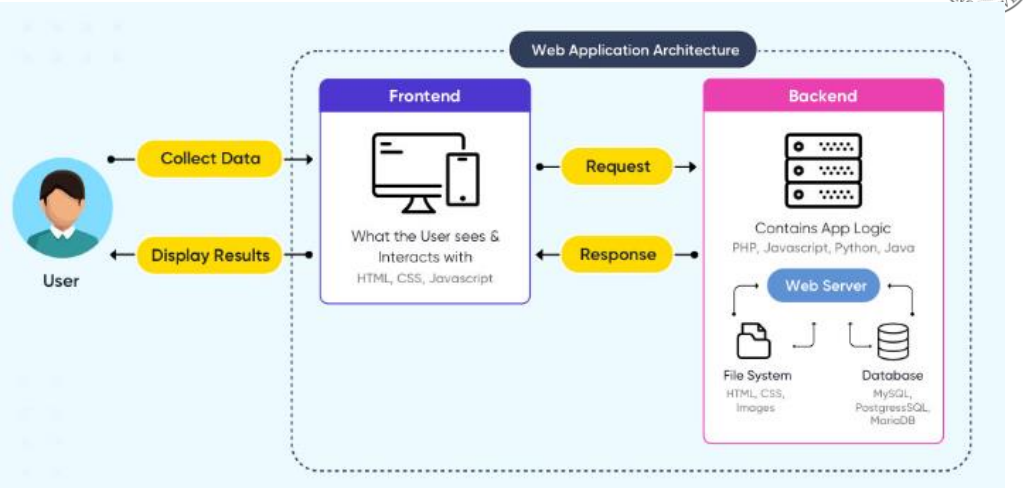
1) PHP
2) MySQL
3) JavaScript, Ajax, jQuery

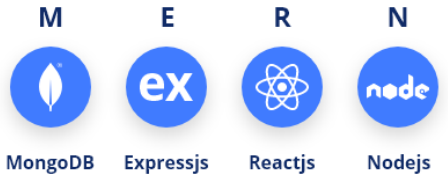
중간 시험

III. MERN Stack

1) Node.js & Express
2) MongoDB
3) React

기말 시험 & Term Project





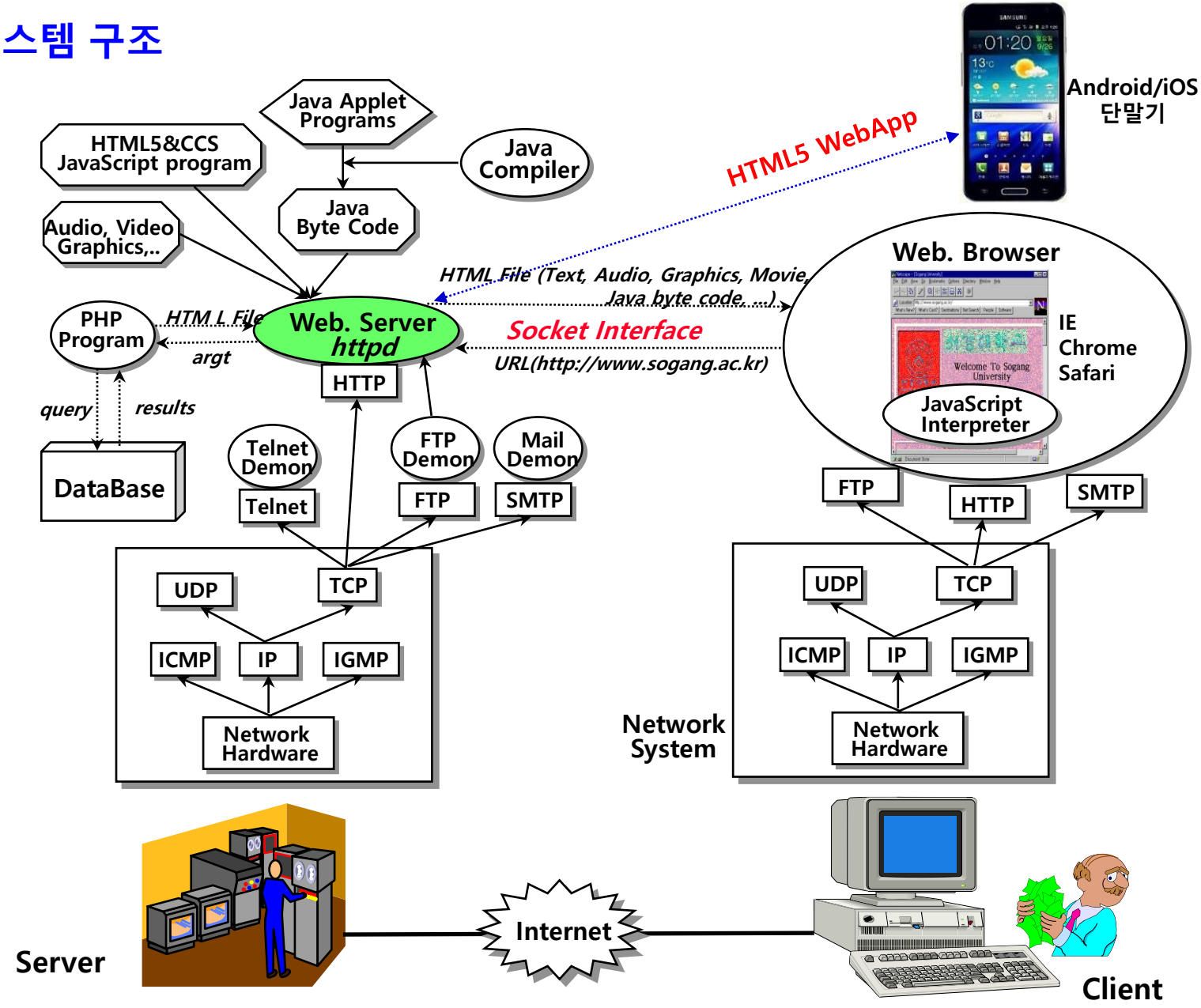


VS



- 조교 : 조윤기 (AS1001, choyoonki@sogang.ac.kr)

- **Web. 시스템 구조**


Java Applet Programs
HTML5&CCS JavaScript program
Java Compiler
Java Byte Code
Audio, Video Graphics,..
HTML5 WebApp
Android/iOS 단말기
Web. Browser
HTML File (Text, Audio, Graphics, Movie, Java byte code, ...)
PHP Program
HTM L File
argt
Web. Server httpd
Socket Interface
URL(http://www.sogang.ac.kr)
IE
Chrome
Safari
JavaScript Interpreter
HTTP
query
results
DataBase
Telnet Demon
Telnet
FTP Demon
FTP
Mail Demon
SMTP
FTP
HTTP
SMTP
UDP
TCP
ICMP
IP
IGMP
Network Hardware
Network System
Server
Internet
Client

LOADING A PAGE IN A BROWSER

representations of resources

Browser

HTML

Other Resources

http://creativecommons.org

HTTP GET

```
http://creativecommons.org
<a><span id="home-button">
</span></a>
<div id="logo">
 <span>
   Creative Commons
 </span>
```

HTTP GET

```
cforms.js
//Collap
String.p
function
  return
this.rep
```

```
creativecommons.css
topbar #home-button{
 position: relative;
 float: lef
 display: b
 height: 40
 width: 15
```

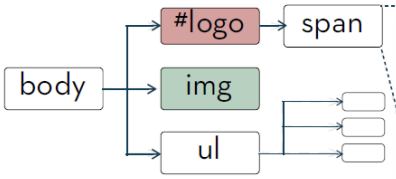
cc-logo.png





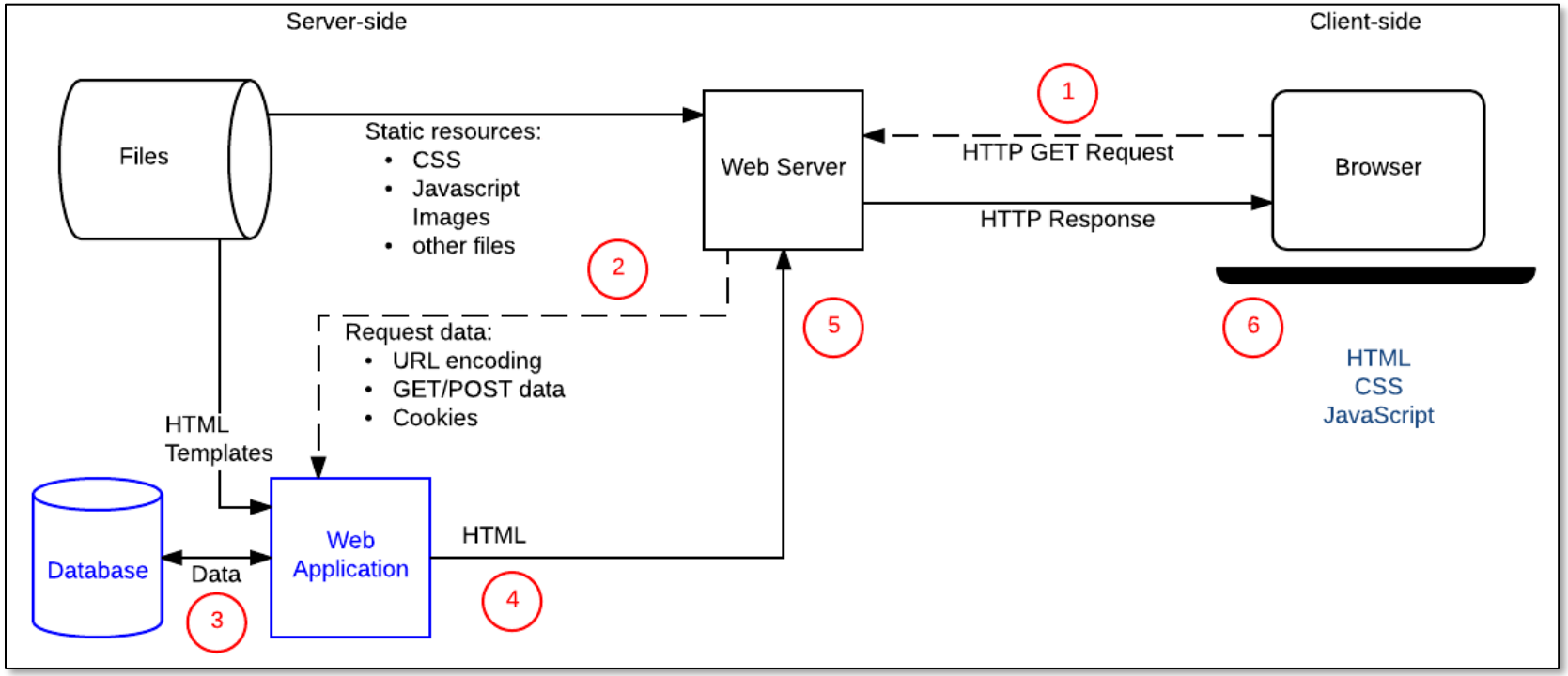
Rendered Page

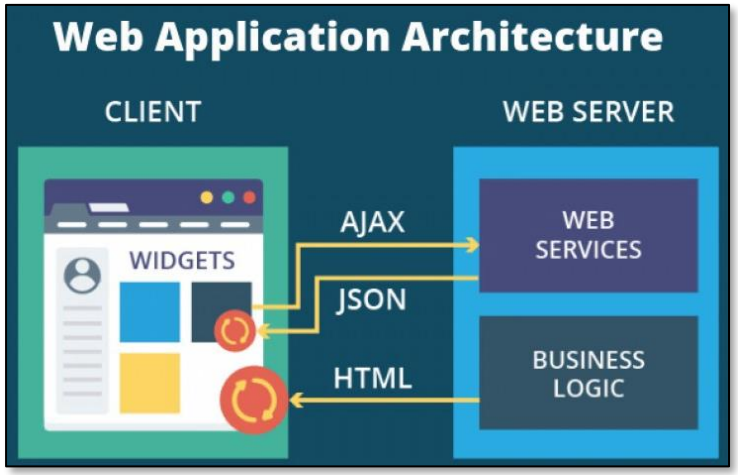
Document Object Model (DOM)

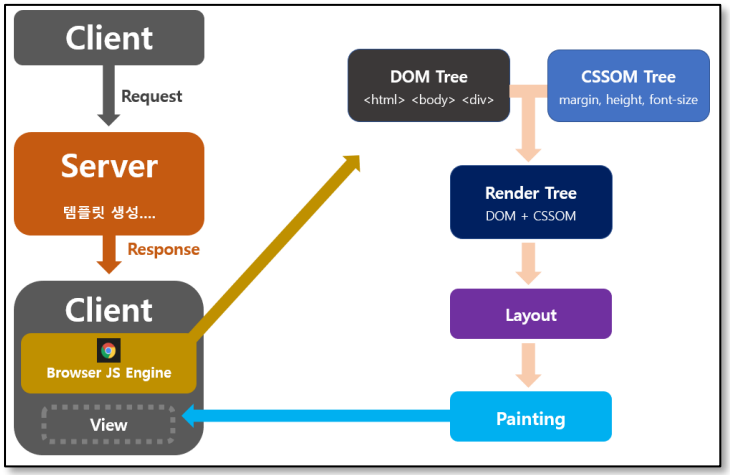


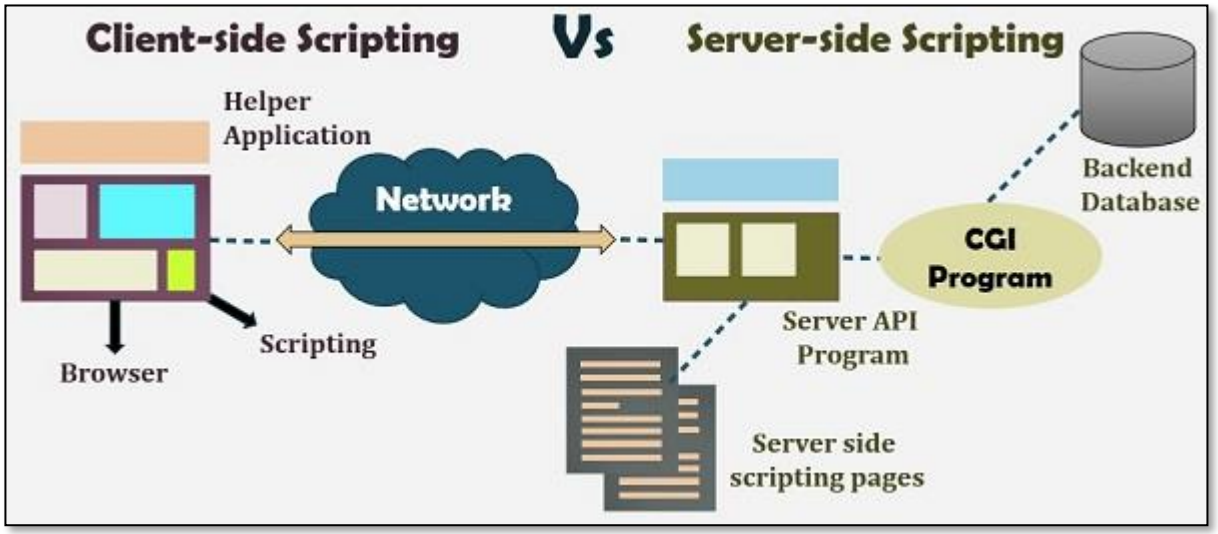

```
topbar span {
  float: left;
  display: block;
  height: 40px;
  width: 150px;
  cursor:
pointer;
  z-index: 1;
  top: 0;
```

1. Web Application Architecture

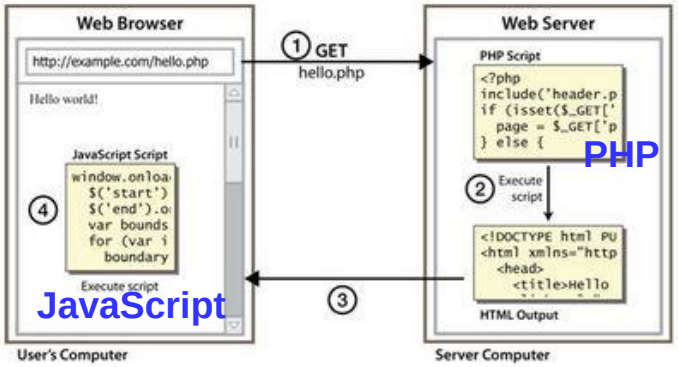






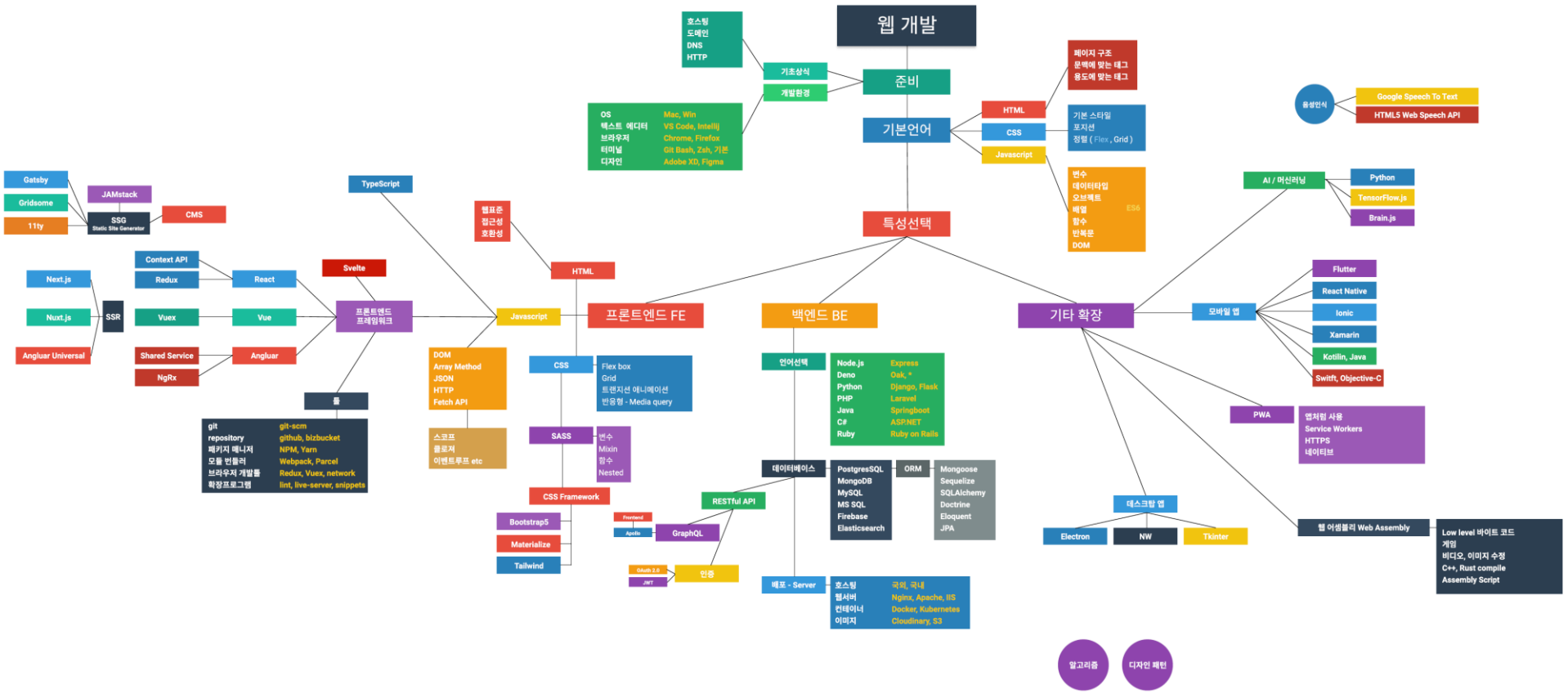


Client-side scripting



- **client-side script**: code runs in browser *after* page is sent back from server
often this code manipulates the page or responds to user actions

• Web Technology Tree



- W3Techs.com

Market Share (24 Aug. 2026, https://w3techs.com/technologies)

Most popular server-side programming languages

© W3Techs.com	usage	change since 1 July 2026
1. PHP	70.2%	-0.6%
2. JavaScript	7.2%	+0.5%
3. Ruby	7.0%	+0.2%
4. Java	5.4%	-0.1%
5. Scala	5.1%	+0.1%

percentages of sites

Most popular client-side programming languages

© W3Techs.com	usage	change since 1 July 2026
1. JavaScript	98.9%	
2. Flash	2.8%	

percentages of sites

Most popular JavaScript libraries

© W3Techs.com	usage	change since 1 July 2026	market share	change since 1 July 2026
1. jQuery	66.5%	-1.5%	86.5%	-0.8%
2. Bootstrap	15.1%	-0.4%	19.7%	-0.2%
3. Underscore	7.1%	-0.2%	9.3%	-0.1%
4. React	6.1%		7.9%	+0.1%
5. Modernizr	5.2%	-0.2%	6.7%	-0.2%

percentages of sites

Most popular web servers

© W3Techs.com	usage	change since 1 July 2026
1. Nginx	31.4%	-0.4%
2. Cloudflare Server	29.6%	+1.1%
3. Apache	22.6%	-0.6%
4. LiteSpeed	14.9%	-0.2%
5. Node.js	7.0%	+0.4%

percentages of sites

Most popular operating systems

© W3Techs.com	usage	change since 1 July 2026
1. Unix	92.0%	+0.3%
2. Windows	8.2%	-0.3%

percentages of sites

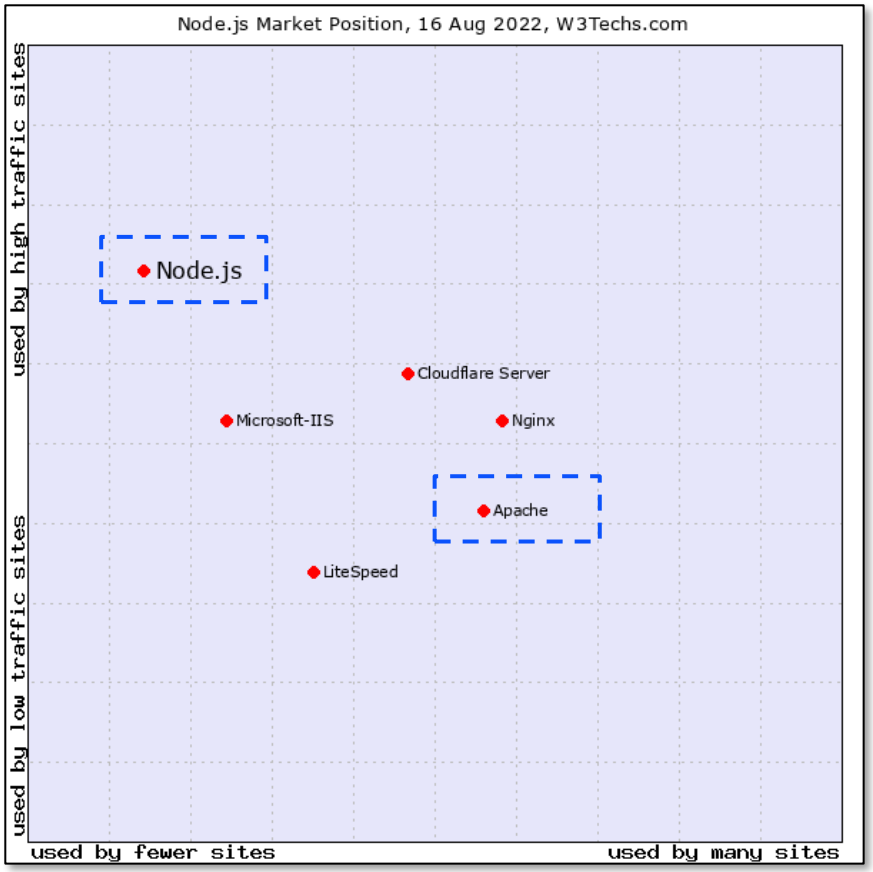
- **Programming languages used in most popular websites**

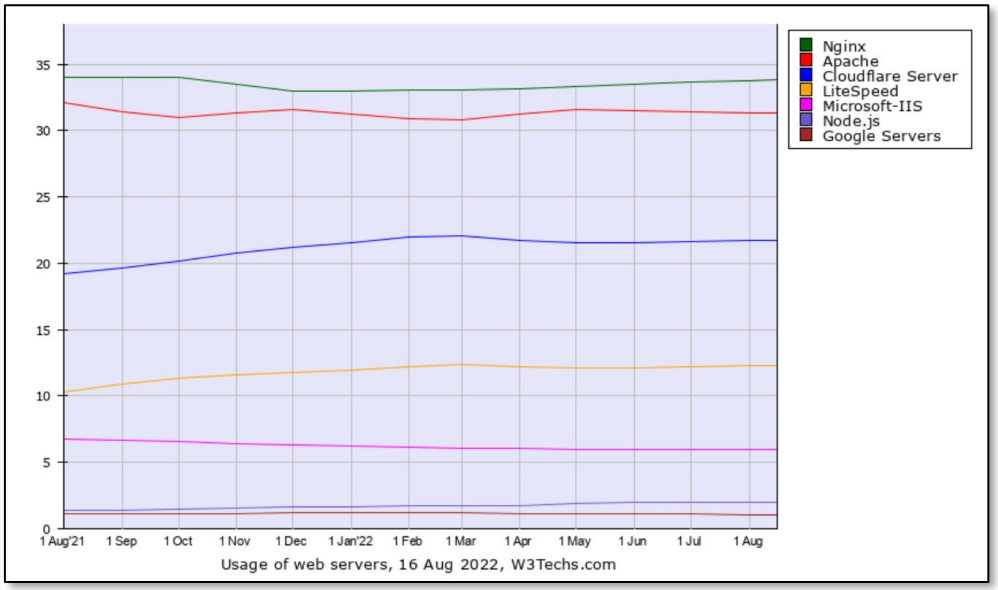
Websites	Popularity (unique visitors per month)[1]	Front-end (Client-side)	Back-end (Server-side)	Database
Google[2]	2,800,000,000	JavaScript, TypeScript	C, C++, Go,[3] Java, Python, Node	Bigtable,[4] MariaDB[5]
Facebook	1,100,000,000	JavaScript, TypeScript, Flow	Hack/HHVM, PHP, Python, C++, Java, Erlang, D,[6] Haskell[7]	MariaDB, MySQL,[8] HBase, Cassandra[9]
YouTube	1,100,000,000	JavaScript, TypeScript	Python, C, C++, Java,[10] Go[11]	Vitess, BigTable, MariaDB[5]
Yahoo	750,000,000	JavaScript	PHP	PostgreSQL, HBase, Cassandra, MongoDB,[12]
Etsy	516,000,000 (Total, not unique)[13]	JavaScript	PHP[14]	MySQL, Redis[15]
Amazon	2,400,000,000[16]	JavaScript	Java, C++, Perl[17]	DynamoDB, RDS/Aurora, Redshift[18]
Wikipedia	475,000,000	JavaScript	PHP	MariaDB[19]
Fandom	315,000,000[20]	JavaScript	PHP	MySQL
X	290,000,000	JavaScript	C++, Java,[21] Scala,[22] Ruby (Ruby on Rails)	MySQL[23]
Bing	285,000,000	JavaScript	C++, C#	Microsoft SQL Server, Cosmos DB
eBay	285,000,000	JavaScript	Java,[24] JavaScript,[25] Scala[26]	Oracle Database

https://en.wikipedia.org/wiki/Programming_languages_used_in_most_popular_websites

언어 등	프레임워크명
JavaScript	React(Facebook, Twitter), Vue.js(LINE, Apple), jQuery,Node.js※
TypeScript	Angular(Google, Microsoft), React, Vue.js, Node.js※
Perl	Catalyst
PHP	CakePHP
JSP	SeeSea, Struts
Python	Django(Instagram)
Ruby	Ruby on Rails(CookPad)
CSS	Bootstrap, Sass※

- ( )는 프레임워크를 이용하고 있는 유명한 사이트나 SNS, ※는 프레임워크라기보다는 개발 환경
- 그 밖에 소스 코드 관리 서비스인 GitHub 등의 이용 및 클라우드 사업자의 PaaS 서비스 이용 등도 들 수 있다
- 개발하고 싶은 것에 대해서 적합한 프레임워크를 선택하면 언어도 필연적으로 정해진다





http://www.sitepoint.com/sitepoint-smackdown-php-vs-node-js/

```php
<?php
echo 'Hello World!';
?>
```

index.php

Node.js

```node
var http = require('http');
http.createServer(function (req, res) {
res.writeHead(200, {'Content-Type': 'text/plain'});
res.end('Hello World!');
}).listen(3000, '127.0.0.1');
```

hello.js

Server-side 프로그래밍도 JavaScript를 이용하자
➔ Node.js

Web Programming

- Framework *vs.* Library

⇔ 구분 기준 : 응용 프로그램의 흐름 주도권을 누가 가지고 있느냐

– **Framework** *(➔프로그램의 주도권: Framework)*

⇔ 코드를 작성하는 기본적인 틀을 제공해서 보다 효율적으로 어플리케이션을 만들 수 있도록 하는 소프트웨어 환경

⇔ 응용 프로그램은 프레임워크에 의해 사용된다.

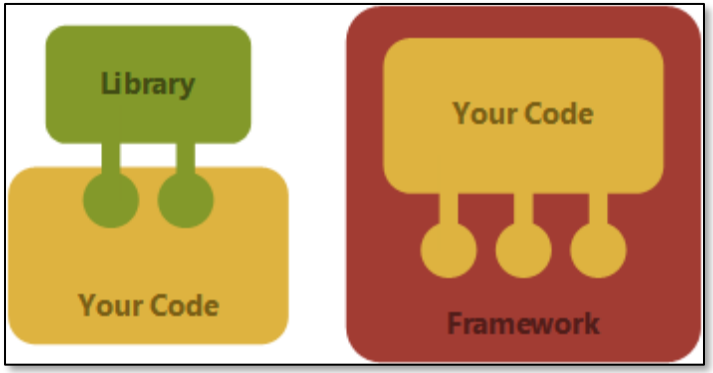
⇔ 키워드: Rule, Logic, Auto

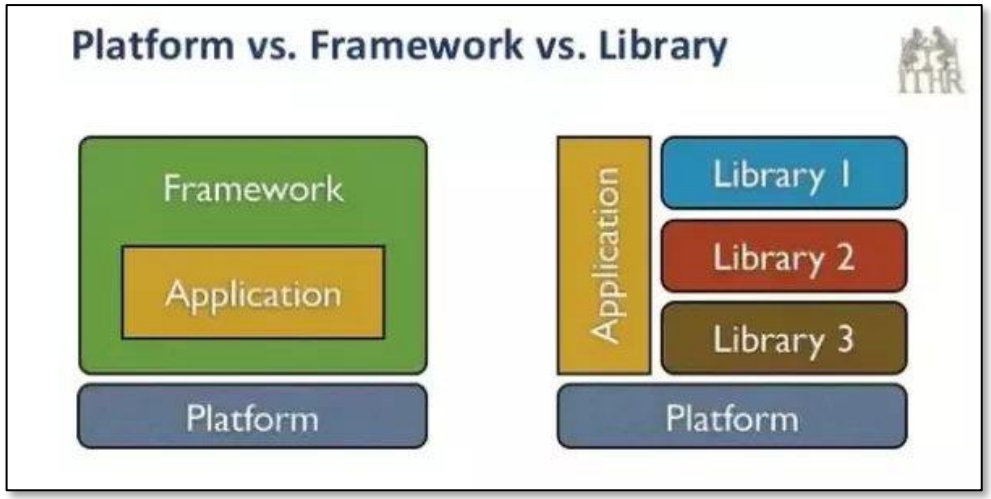
– **Library** *(➔프로그램의 주도권 : 응용 프로그램)*

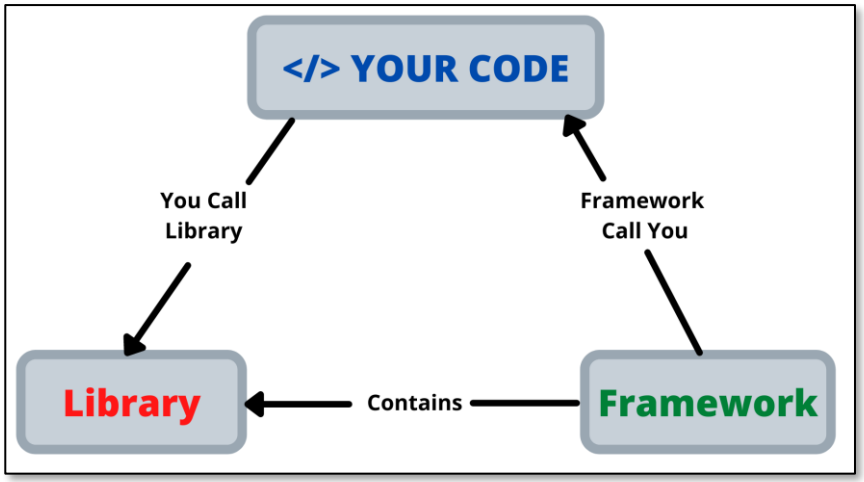
⇔ 어플리케이션을 만들 때 필요한 자원(기능: 함수)의 모임

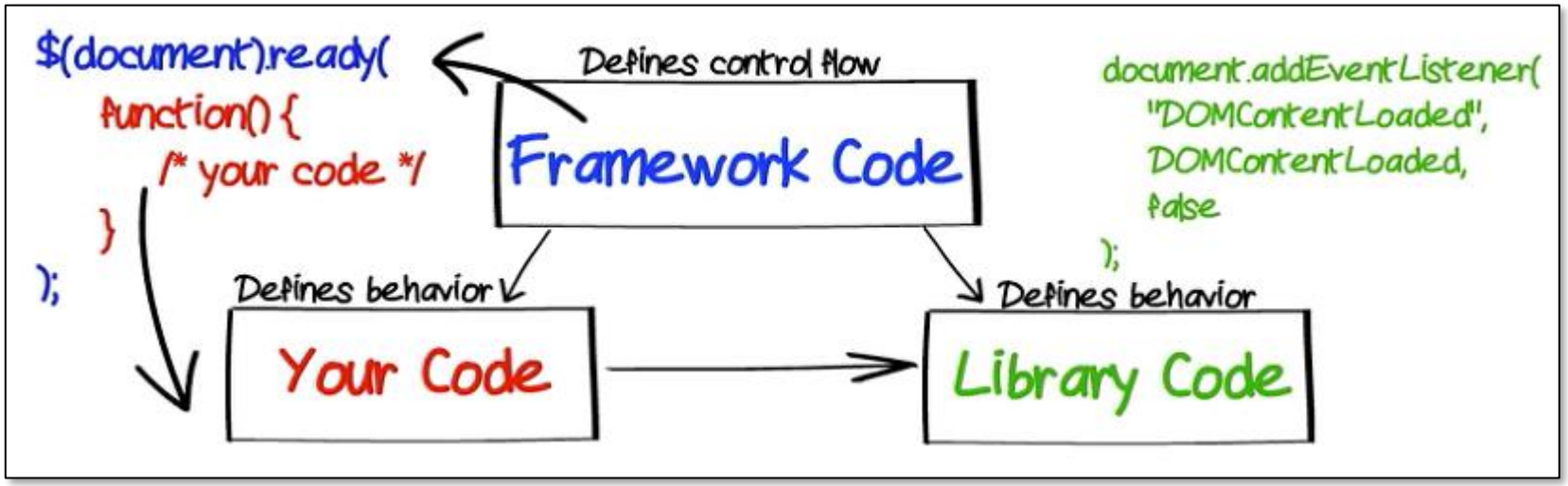
⇔ 응용 프로그램이 라이브러리를 사용한다.

⇔ 키워드: Freedom, Custom

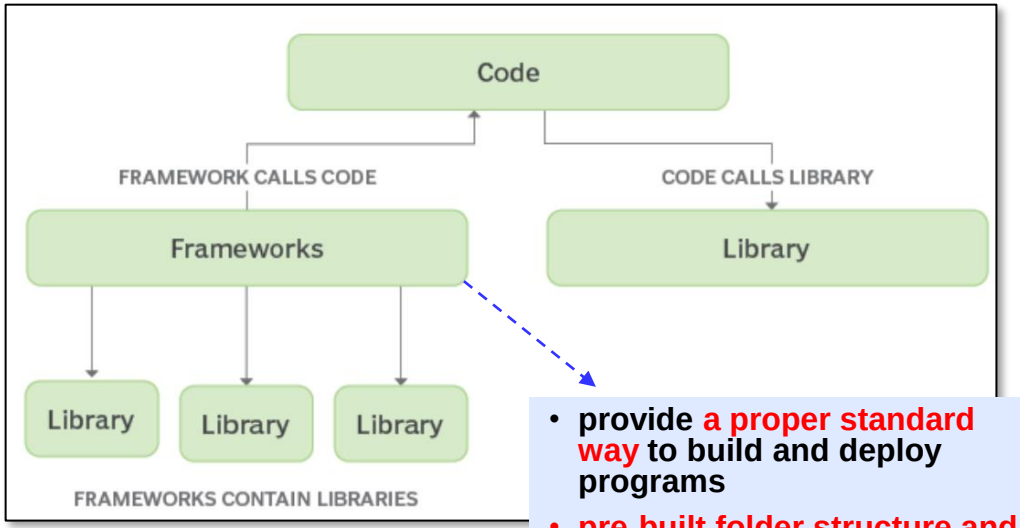




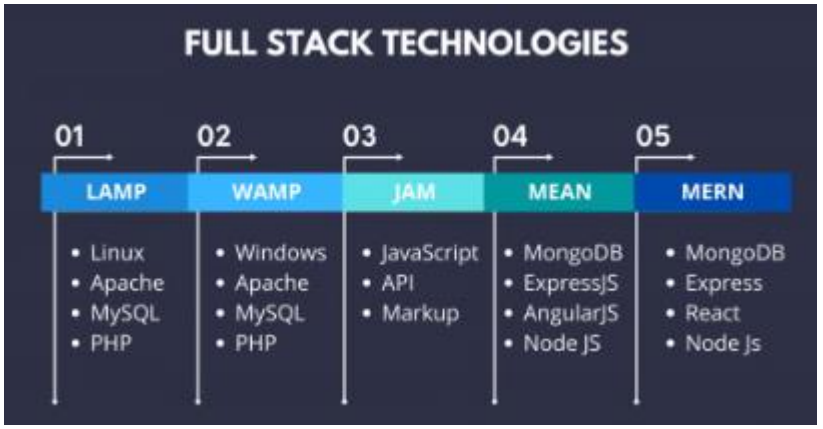




LIBRARY	VS	FRAMEWORK
Library is a set of reusable functions used by computer program		It is a piece of code that dictates the artitechture of our project
They are important in program linking and binding process		They provide a proper standard way to build and deploy programs
Code calls to library		Framework calls the code
Jquery is the javascript library		Angular js is javascript baesd framework



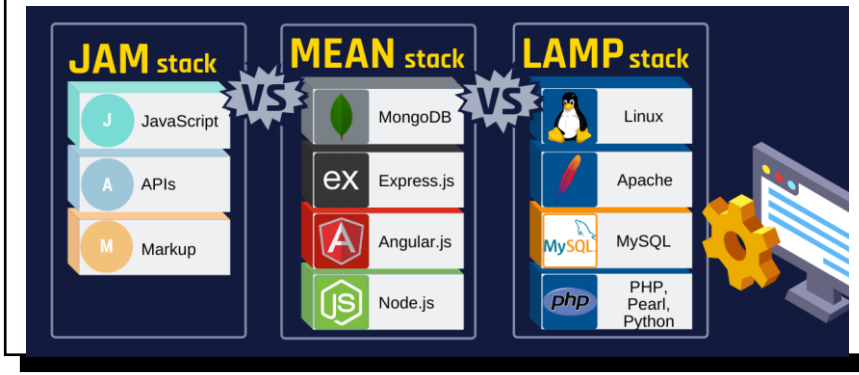
- provide a proper standard way to build and deploy programs
- pre-built folder structure and file names





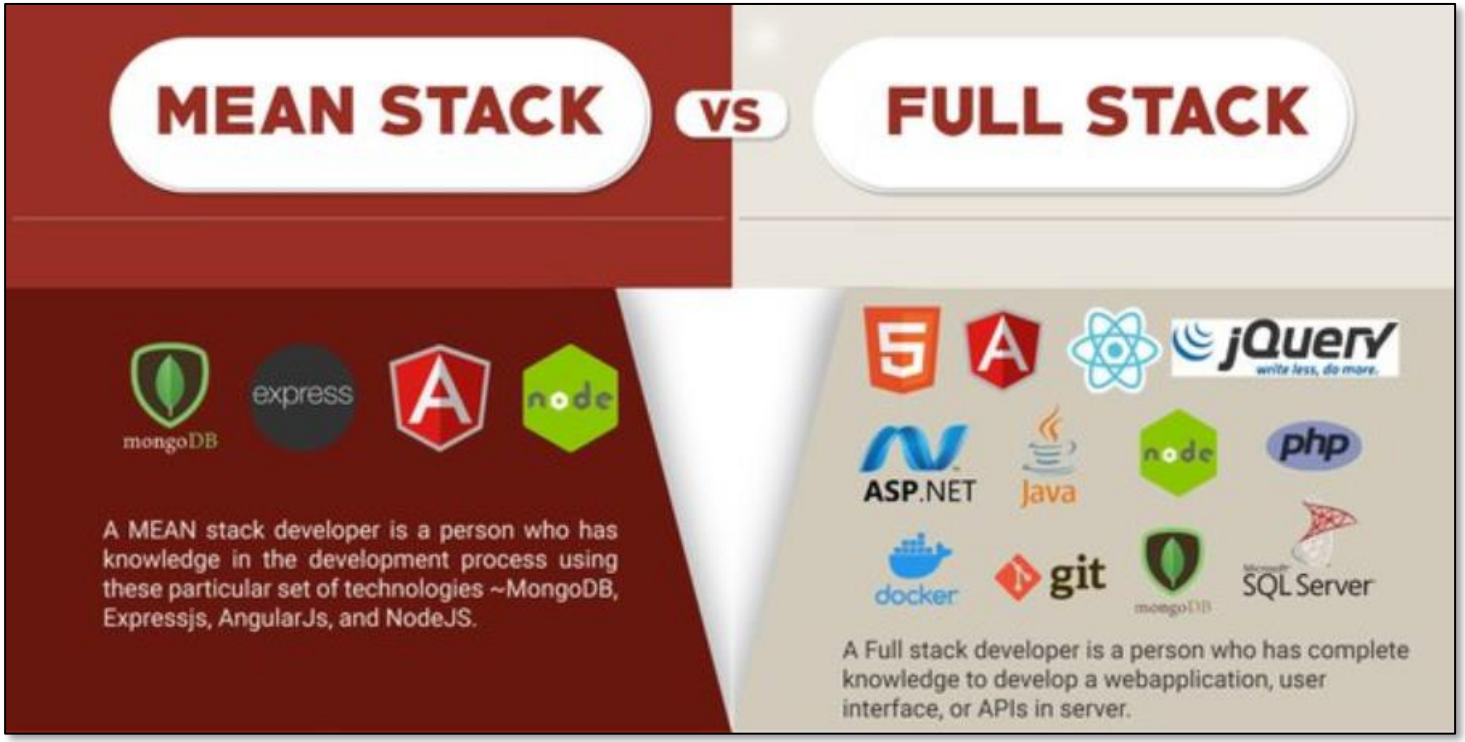
Framework **Library**





	MEAN Stack	LAMP Stack
Operating System	Cross-platform	Linux OS only
Database	MongoDB, a 'Non-Relational' Database	MySQL, a 'Relational' Database
Programming Language(s)	Frontend—JavaScript, Backend—PHP	JavaScript, backend and frontend
Structure	Slower & limited due to its blocking structure	Faster & Scalable due to its non-blocking structure.
Technologies	MongoDB, Express.js, AngularJS, Node.js	Linux, Apache, MySQL, PHP/Ruby/Python
Mainstream Backers	Google, IBM, Samsung	Oracle, Zend, Linux Foundation

- **What is difference between MEAN stack developer and FULL stack developer?**
  - ✓ A Full-stack developer is a web developer or engineer who works with ***both the front and back ends of a website or application***—meaning they can tackle projects that involve databases, building user-facing websites, or even work with clients during the planning phase of projects.
  - ✓ A MEAN stack developer is someone who has knowledge in ***some particular areas alone.*** MEAN stack development refers to the development process that falls within these particular set of technologies MongoDB, Expressjs, AngularJs, Node.js



Full-Stack Developer Skills Breakdown

FRONT-END

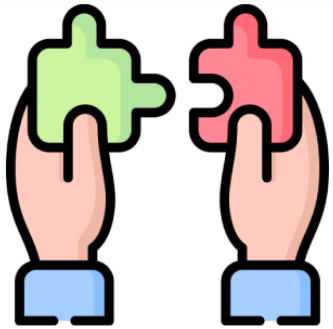
Basics
HTML
CSS
Javascript

Frameworks
React
Vue
Angular
Webpack

Styles
Bootstrap
Material UI

BACK-END

Technology
PHP
Node
Ruby on Rails
Python
Java
ASP.Net



DATABASE

RDBMS
MSSQL
MYSQL
Postgres

NoSQL
Mongo
Casandra
CouchDB
Elasticsearch

Message Queues
Kafka
SQS
ZeroMQ
RabbitMQ

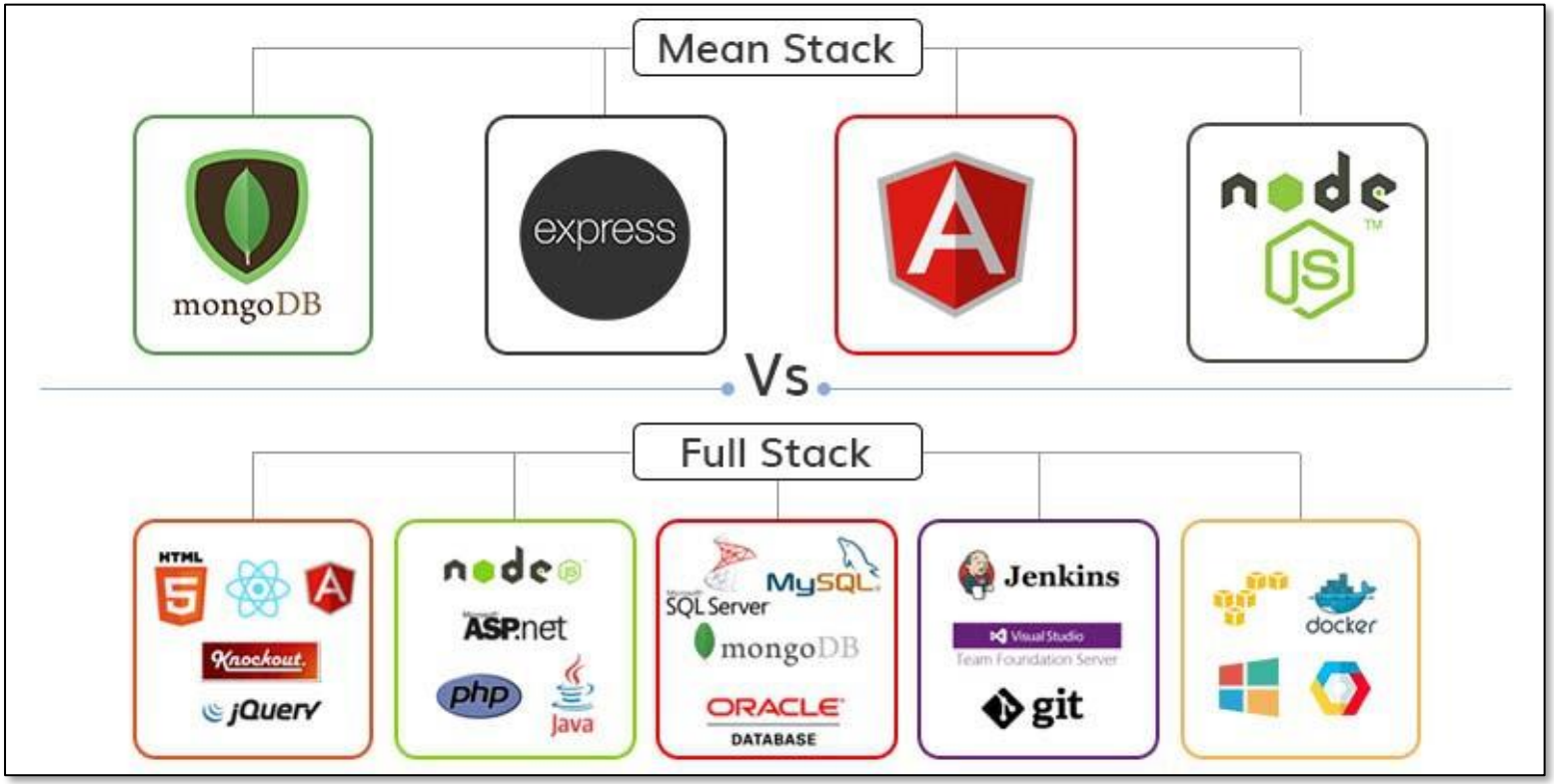
DEV OPS

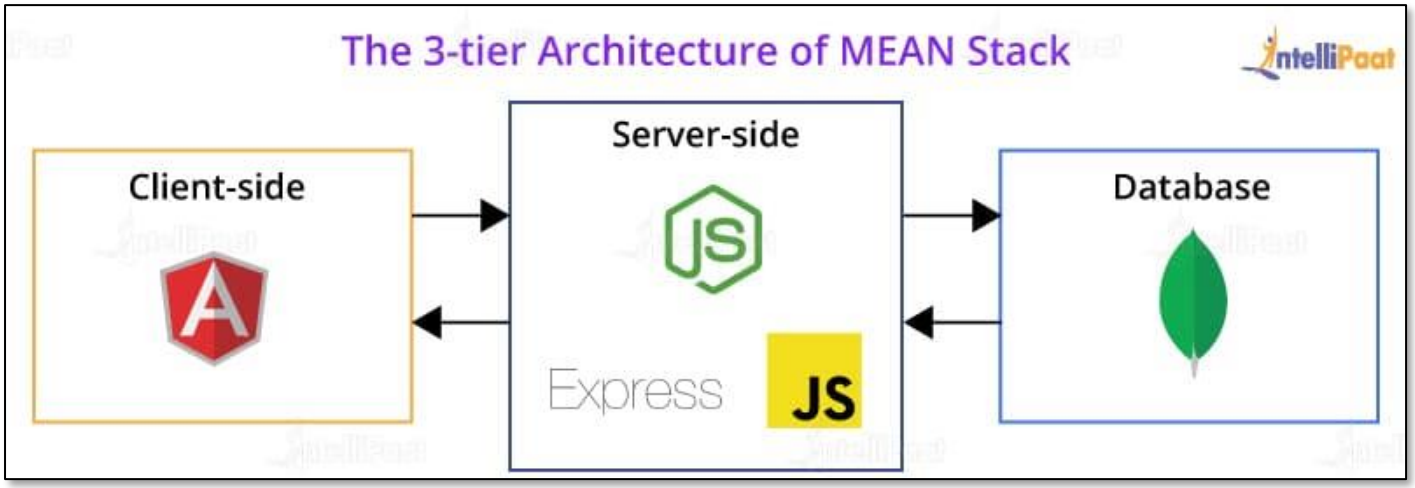
Infrastructure
AWS
Azure
NGINX
ELK

Virtualization
Docker
VMWare
Kubernetes
Vagrant

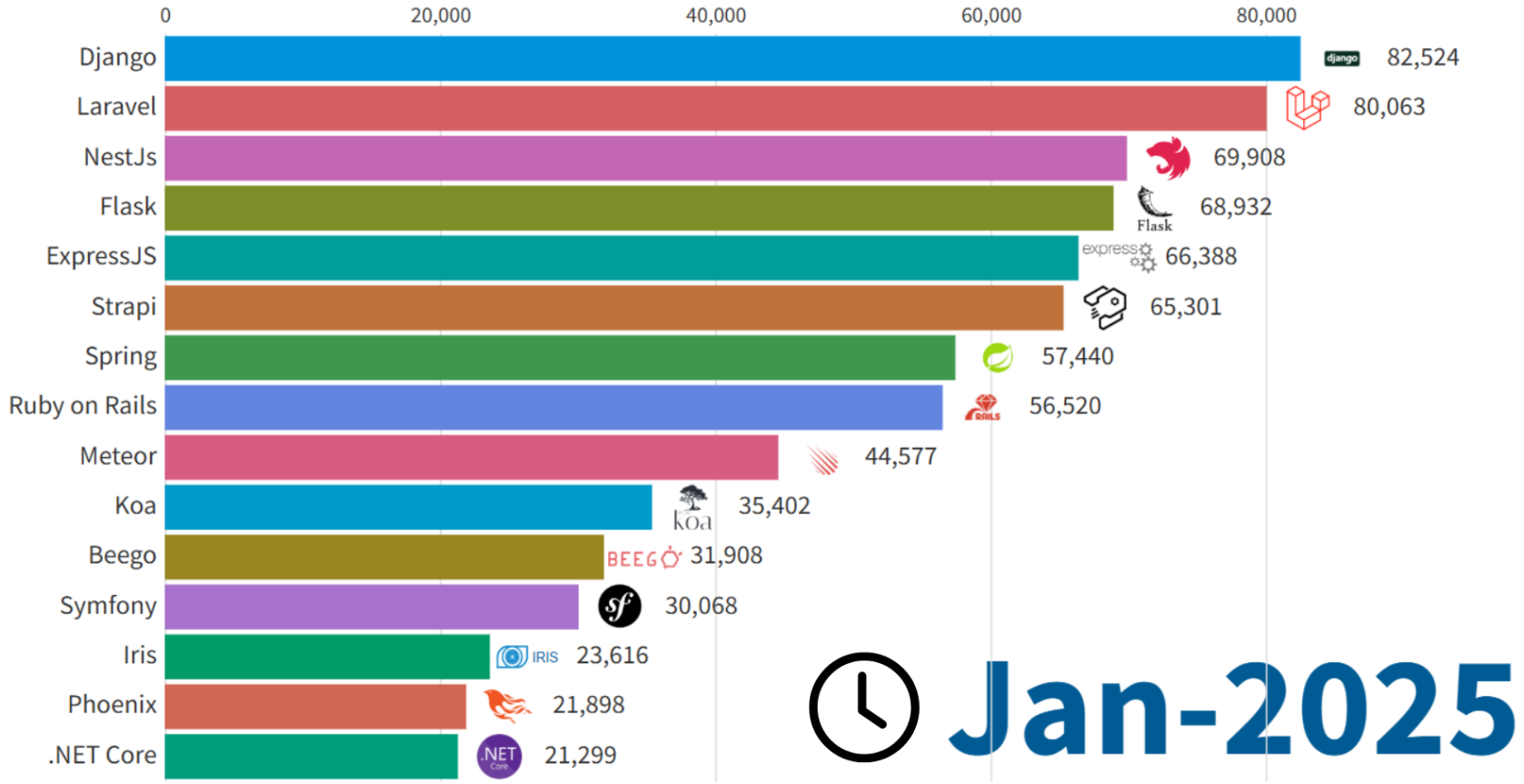
Automation
Ansible
Chef
Jenkins

Adaface


Mean Stack
mongoDB
express
node
JS
Vs
Full Stack
HTML
Knockout.
jQuery
node
ASP.net
php
Java
MySQL
SQL Server
mongoDB
ORACLE
DATABASE
Jenkins
Visual Studio
Team Foundation Server
git
docker


The 3-tier Architecture of MEAN Stack
IntelliPaat
Client-side
Server-side
Express
JS
Database

Most Popular Backend Frameworks



https://statisticsanddata.org/data/most-popular-backend-frameworks-2012-2025/

2026–27 개설: Winter / Spring, 3 units

1. CS 142 — Web Applications ★★★★★

Stanford CS의 대표적인 **Web Application 전공 과목**입니다. Stanford Bulletin에서도 Computer Science 학사(BS) 과정에 포함되는 과목으로 명시되어 있습니다.

Stanford Univers... +1

2026–27 개설: Winter / Spring, **3 units**

주요 내용은 다음과 같습니다.

- HTML / CSS
- DOM
- JavaScript 및 JavaScript Framework
- Client-side programming
- Server-side JavaScript
- Session 관리
- Database
- Web Security
- Application Scalability
- Web Application Deployment

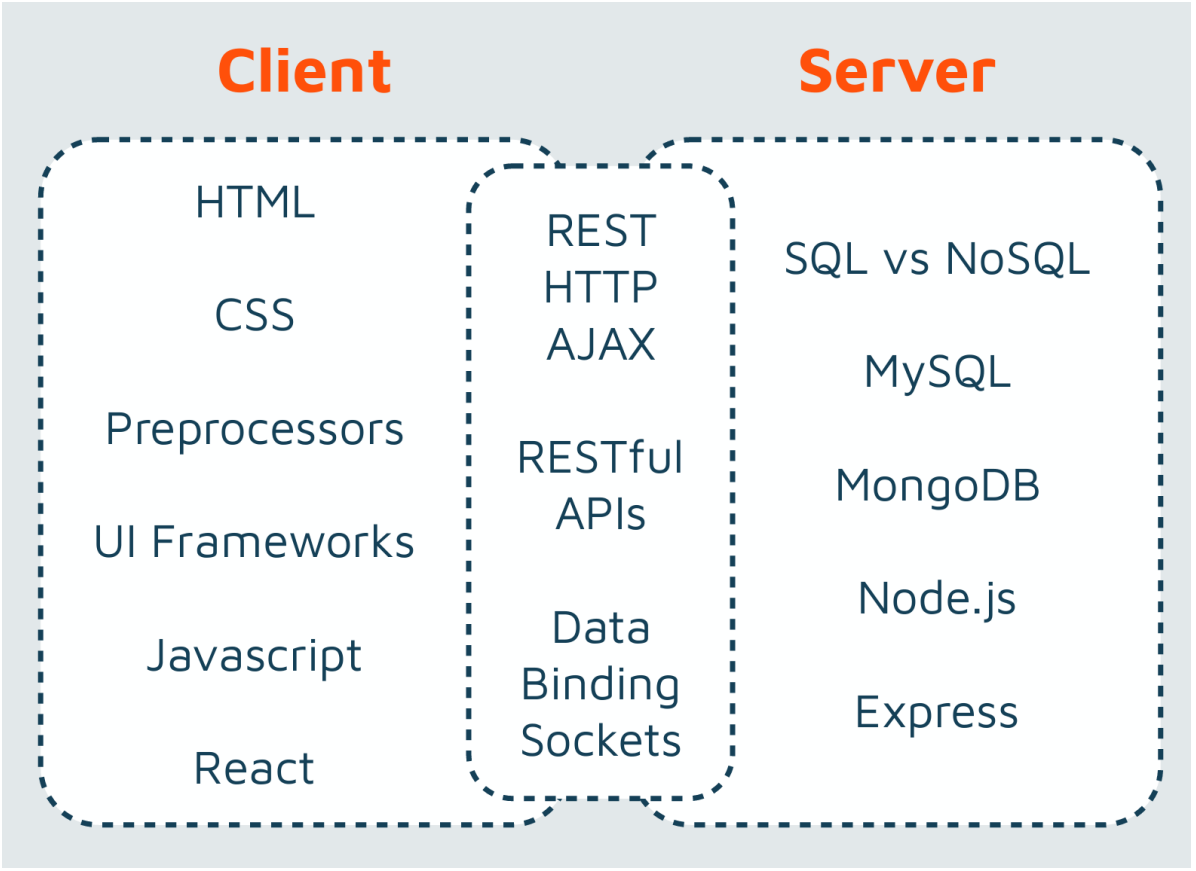
https://bulletin.stanford.edu/courses/2085161/tab-aoYks?utm_source=chatgpt.com

https://chatgpt.com/c/6a8b87ac-1094-83e8-8735-124721a86c87

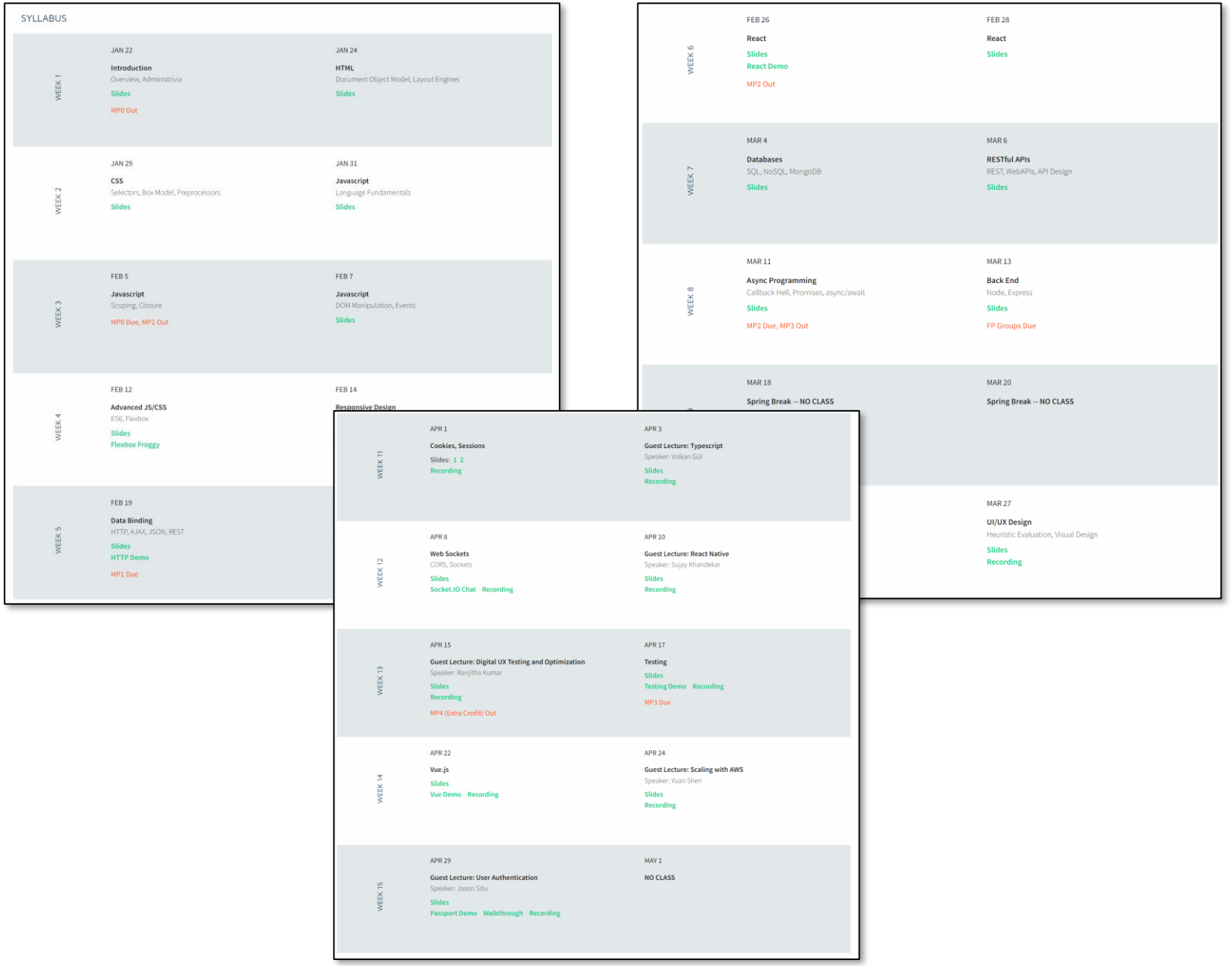
UIUC

The Art and Science of Web Programming

CS498RK SPRING 2020 UIUC WEDNESDAY AND FRIDAY 2PM-3.15PM DCL 1310

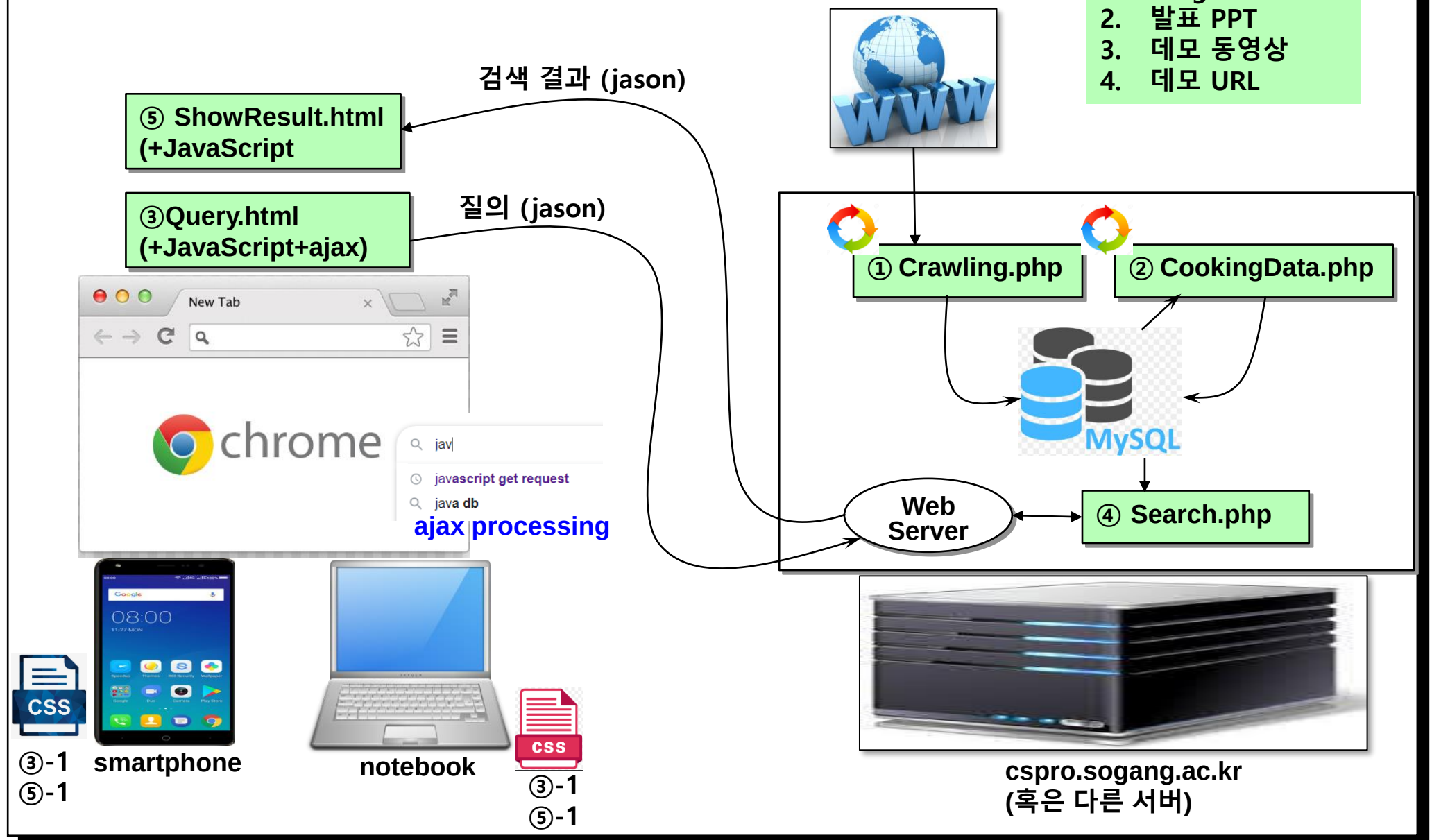


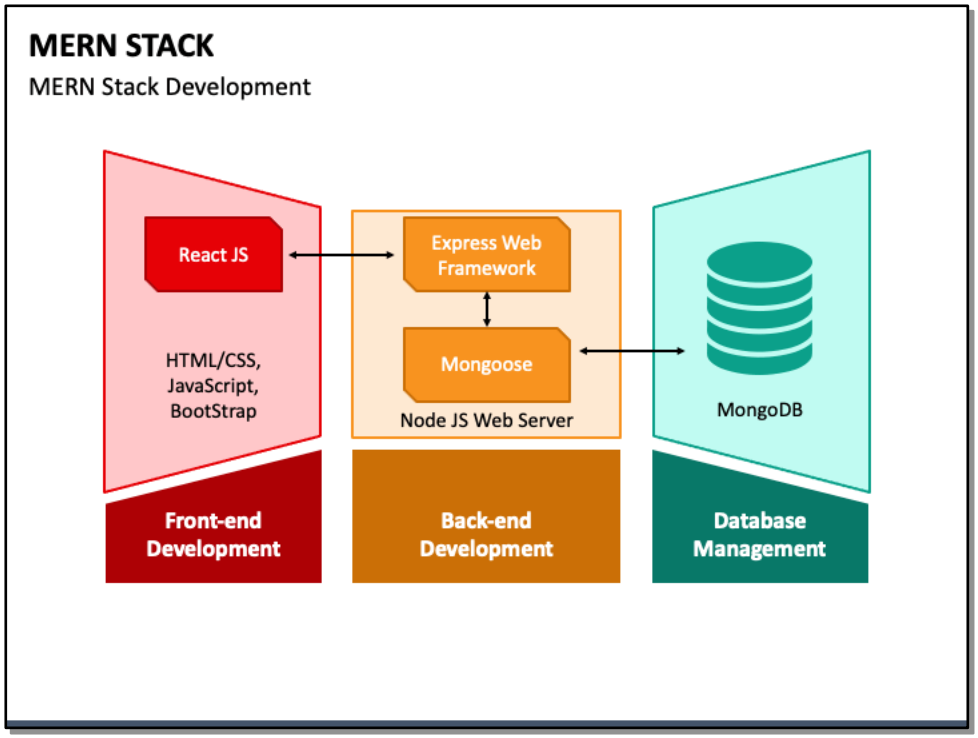
https://uiuc-web-programming.gitlab.io/sp20/

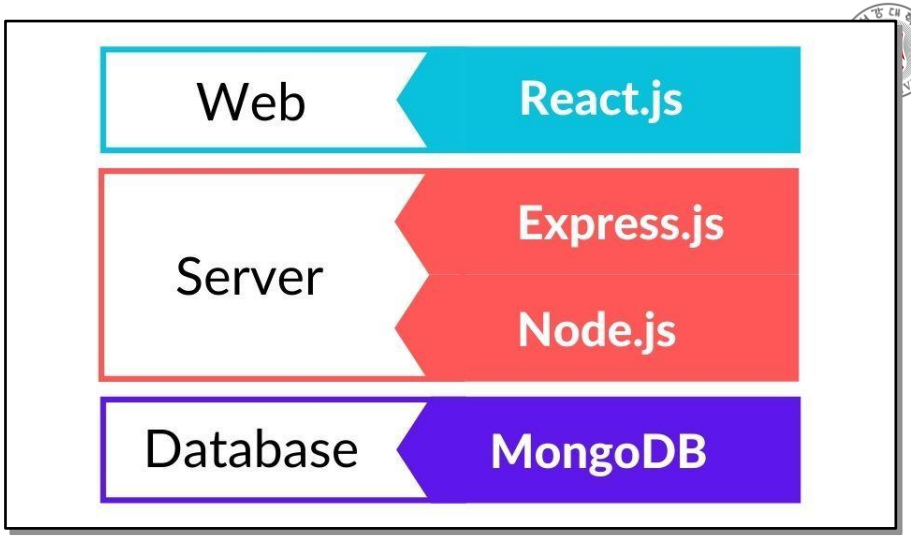

SYLLABUS
WEEK 1
JAN 22
Introduction
Overview, Administrivia
Slides
MP0 Out
JAN 24
HTML
Document Object Model, Layout Engines
Slides
WEEK 2
JAN 29
CSS
Selectors, Box Model, Preprocessors
Slides
JAN 31
Javascript
Language Fundamentals
Slides
WEEK 3
FEB 5
Javascript
Scoping, Closure
MP0 Due, MP1 Out
FEB 7
Javascript
DOM Manipulation, Events
Slides
WEEK 4
FEB 12
Advanced JS/CSS
ES6, Flexbox
Slides
Flexbox Froggy
FEB 14
Responsive Design
WEEK 5
FEB 19
Data Binding
HTTP, AJAX, JSON, REST
Slides
HTTP Demo
MP1 Due
WEEK 6
FEB 26
React
Slides
React Demo
MP2 Out
FEB 28
React
Slides
WEEK 7
MAR 4
Databases
SQL, NoSQL, MongoDB
Slides
MAR 6
RESTful APIs
REST, WebAPIs, API Design
Slides
WEEK 8
MAR 11
Async Programming
Callback Hell, Promises, async/await
Slides
MP2 Due, MP3 Out
MAR 13
Back End
Node, Express
Slides
FP Groups Due
MAR 18
Spring Break -- NO CLASS
MAR 20
Spring Break -- NO CLASS
MAR 27
UI/UX Design
Heuristic Evaluation, Visual Design
Slides
Recording
WEEK 11
APR 1
Cookies, Sessions
Slides: 1 2
Recording
APR 3
Guest Lecture: Typescript
Speaker: Volkan Gül
Slides
Recording
WEEK 12
APR 8
Web Sockets
CORS, Sockets
Slides
Socket.IO Chat Recording
APR 10
Guest Lecture: React Native
Speaker: Sujay Khandekar
Slides
Recording
WEEK 13
APR 15
Guest Lecture: Digital UX Testing and Optimization
Speaker: Ranjitha Kumar
Slides
Recording
MP4 (Extra Credit) Out
APR 17
Testing
Slides
Testing Demo Recording
MP3 Due
WEEK 14
APR 22
Vue.js
Slides
Vue Demo Recording
APR 24
Guest Lecture: Scaling with AWS
Speaker: Yuan Shen
Slides
Recording
WEEK 15
APR 29
Guest Lecture: User Authentication
Speaker: Jason Situ
Slides
Passport Demo Walkthrough Recording
MAY 1
NO CLASS

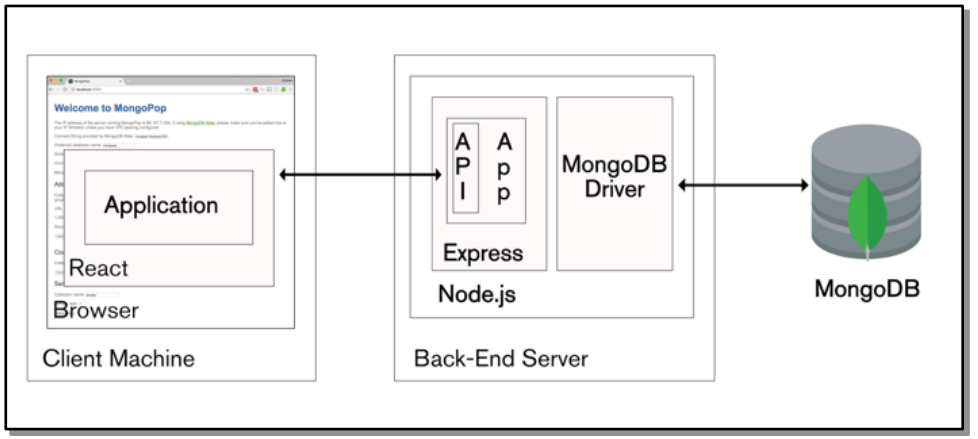
• Term Project (LAMP Stack, MERN Stack)

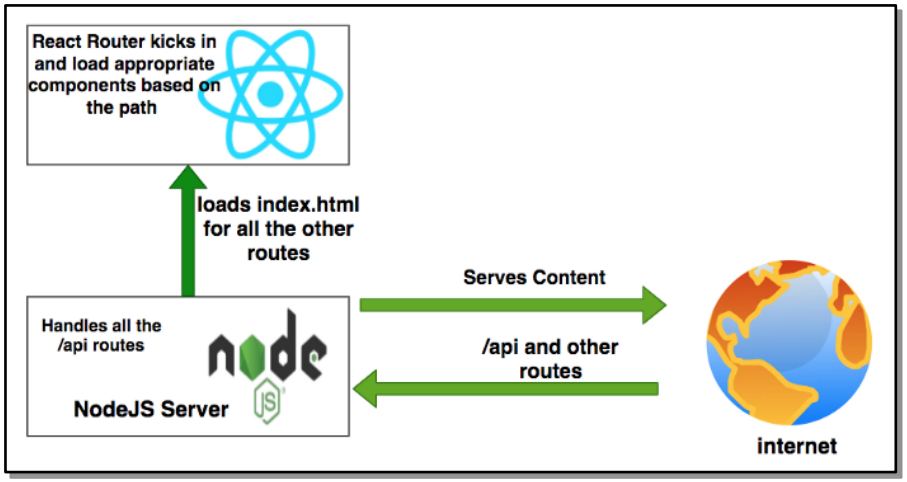
<제출할 것>
1. Program source
2. 발표 PPT
3. 데모 동영상
4. 데모 URL



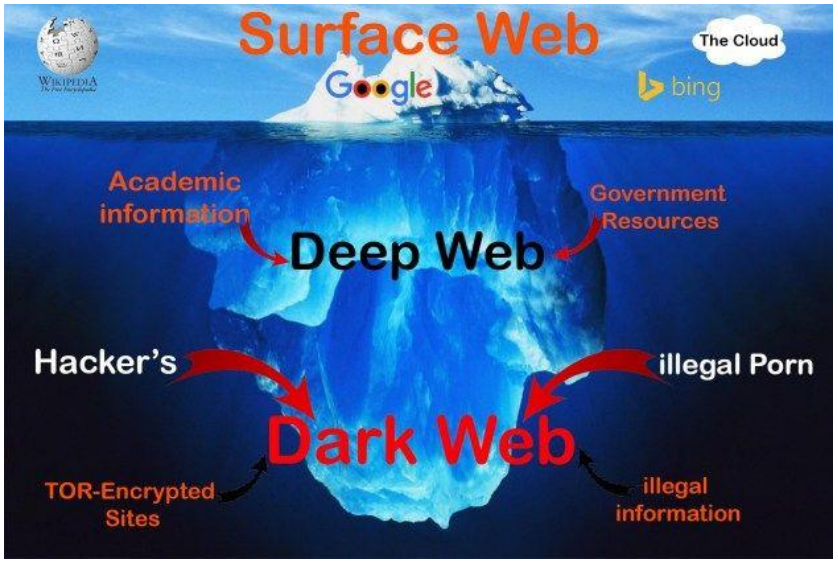

MERN STACK
MERN Stack Development
React JS
HTML/CSS, JavaScript, BootStrap
Express Web Framework
Mongoose
Node JS Web Server
MongoDB
Front-end Development
Back-end Development
Database Management

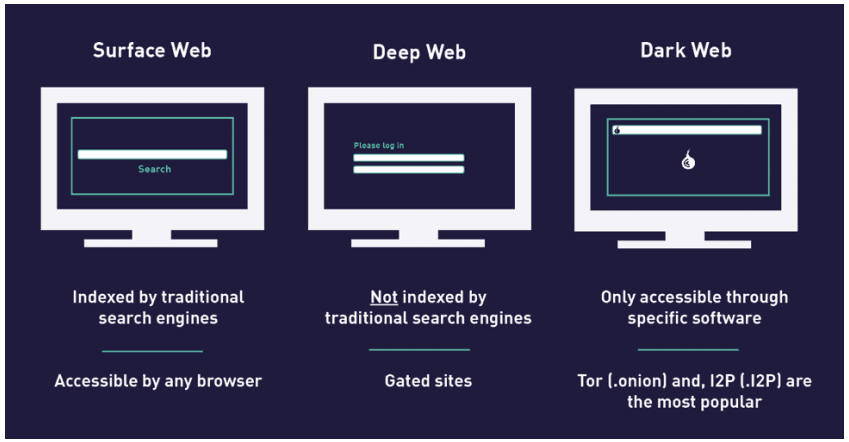

Web
React.js
Server
Express.js
Node.js
Database
MongoDB

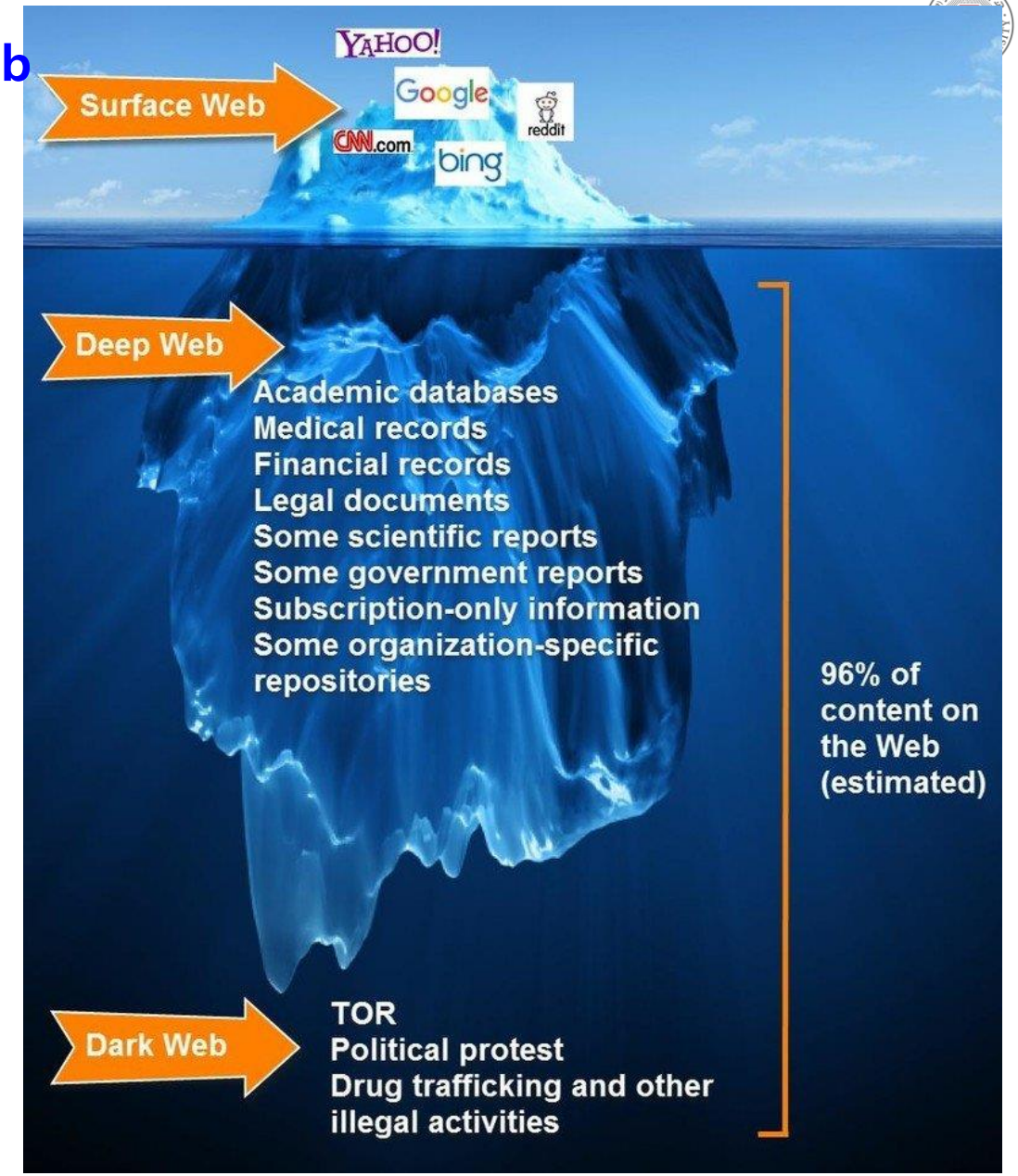

Welcome to MongoPop
Application
React
Browser
Client Machine
API
App
Express
Node.js
MongoDB Driver
Back-End Server
MongoDB

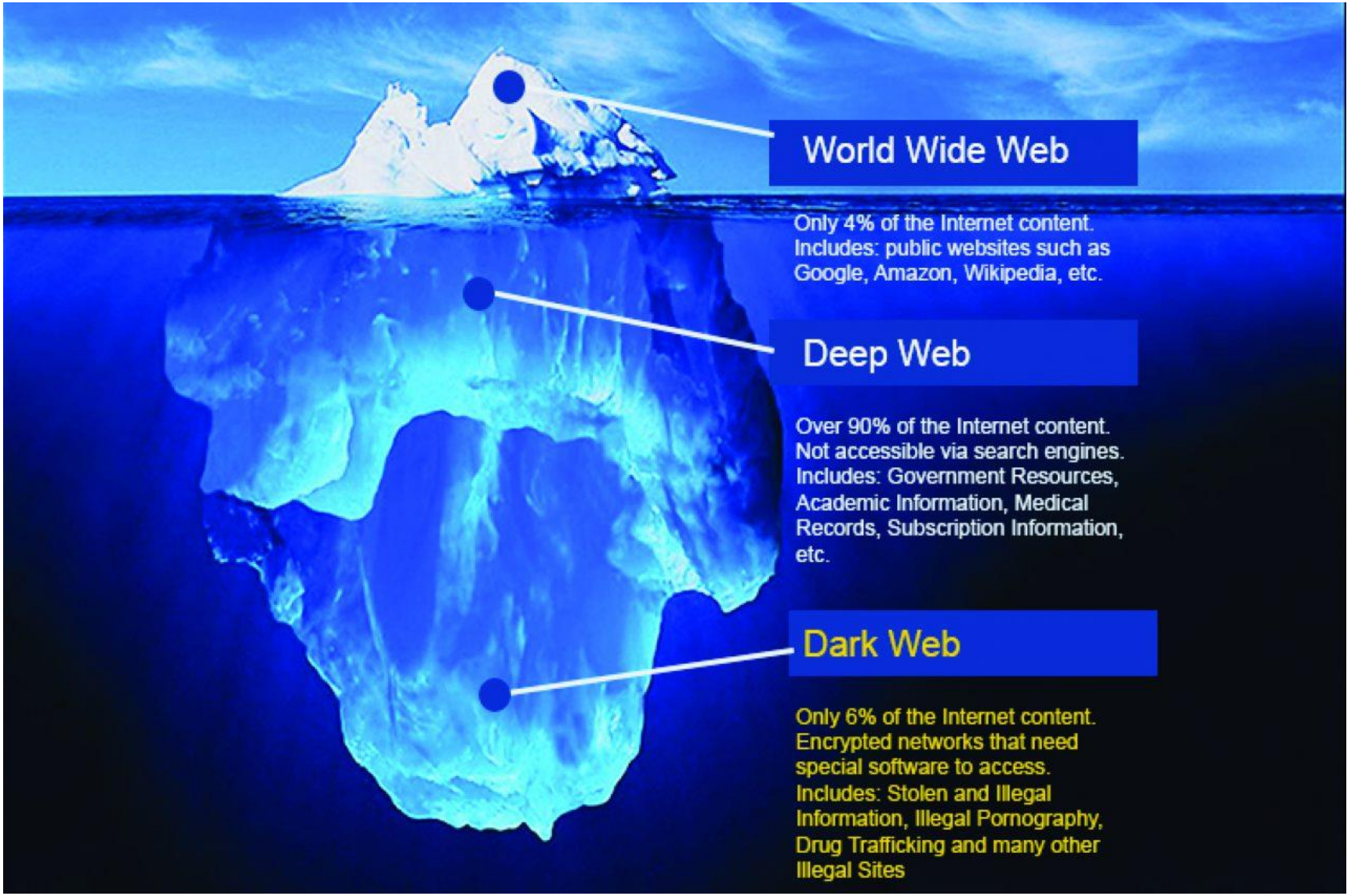

React Router kicks in and load appropriate components based on the path
loads index.html for all the other routes
Handles all the /api routes
NodeJS Server
Serves Content
/api and other routes
internet

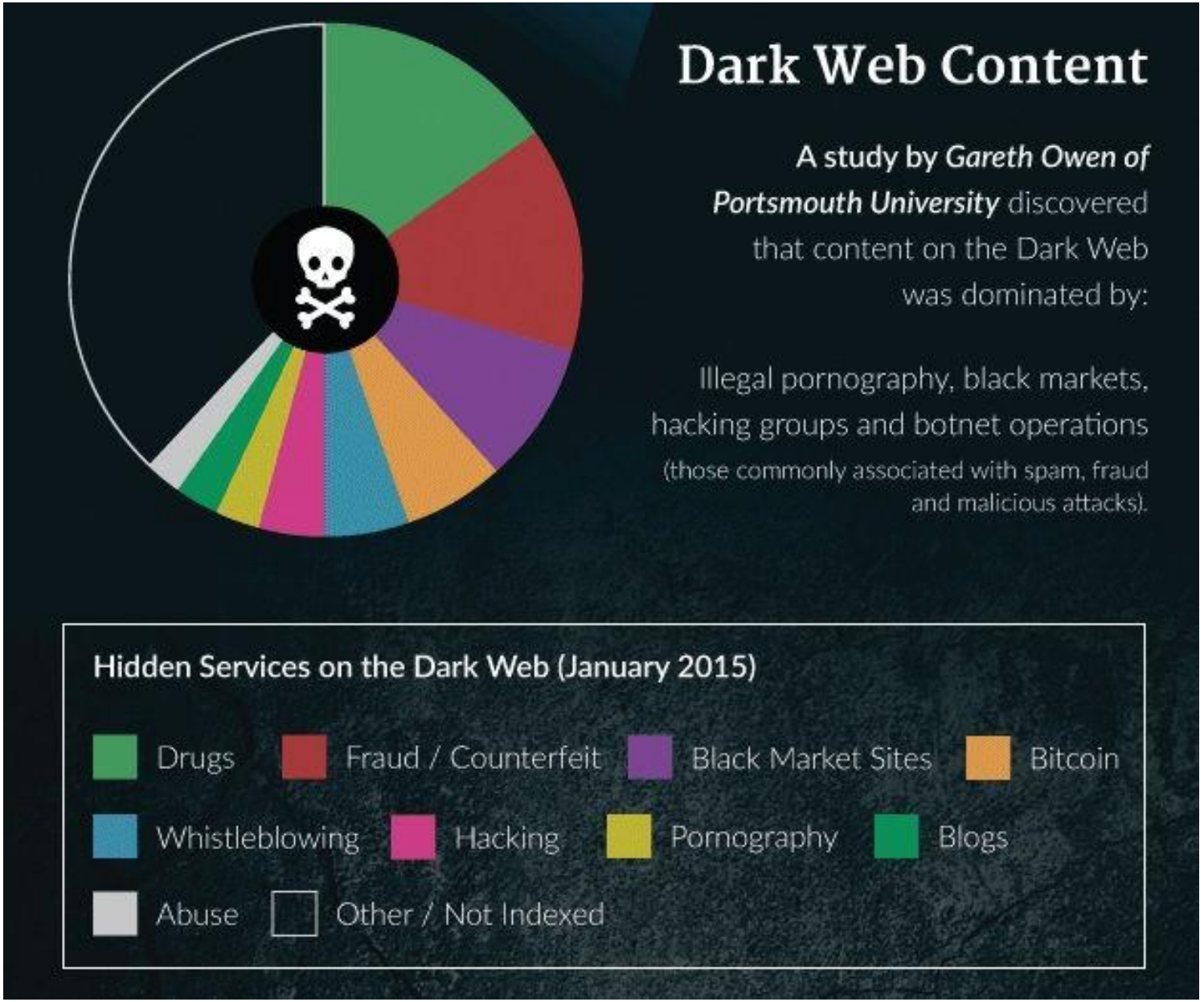
<참고 1> Web, Deep Web, Dark Web








World Wide Web
Only 4% of the Internet content.
Includes: public websites such as Google, Amazon, Wikipedia, etc.
Deep Web
Over 90% of the Internet content.
Not accessible via search engines.
Includes: Government Resources, Academic Information, Medical Records, Subscription Information, etc.
Dark Web
Only 6% of the Internet content.
Encrypted networks that need special software to access.
Includes: Stolen and Illegal Information, Illegal Pornography, Drug Trafficking and many other Illegal Sites


Dark Web Content
A study by Gareth Owen of Portsmouth University discovered that content on the Dark Web was dominated by:
Illegal pornography, black markets, hacking groups and botnet operations
(those commonly associated with spam, fraud and malicious attacks).
Hidden Services on the Dark Web (January 2015)
Drugs
Fraud / Counterfeit
Black Market Sites
Bitcoin
Whistleblowing
Hacking
Pornography
Blogs
Abuse
Other / Not Indexed

The 20 Best Dark Web Sites: Dark Web Links Worth Visiting



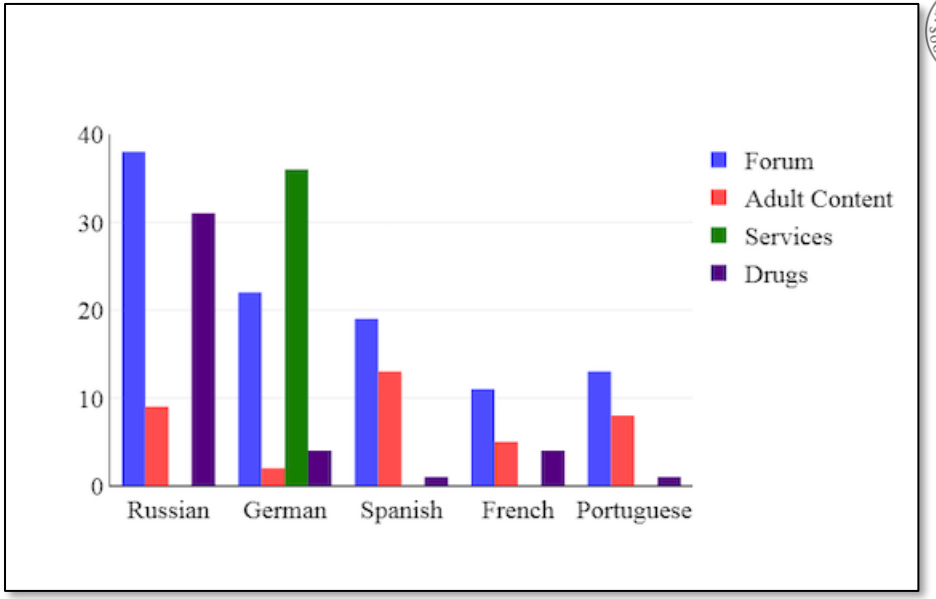
Tove Marks | Reading time: 17 minutes | Update: 07-26-2022



The Best Sites on the Dark Web: A Quick Summary

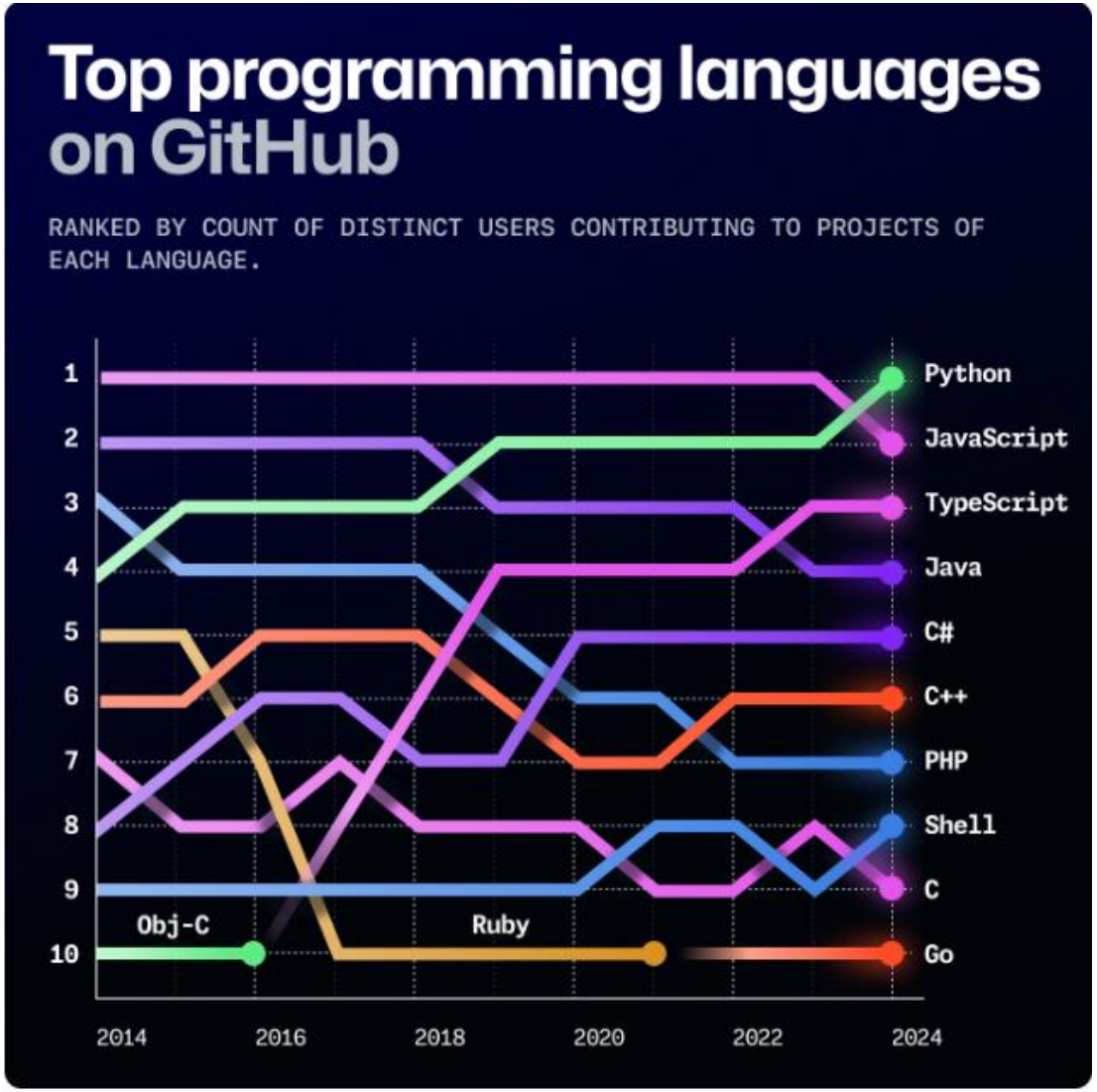
It's hard to tell how many dark web sites there are really out there, but a lot of people agree on the most worthy ones to visit. According to our research, these include:

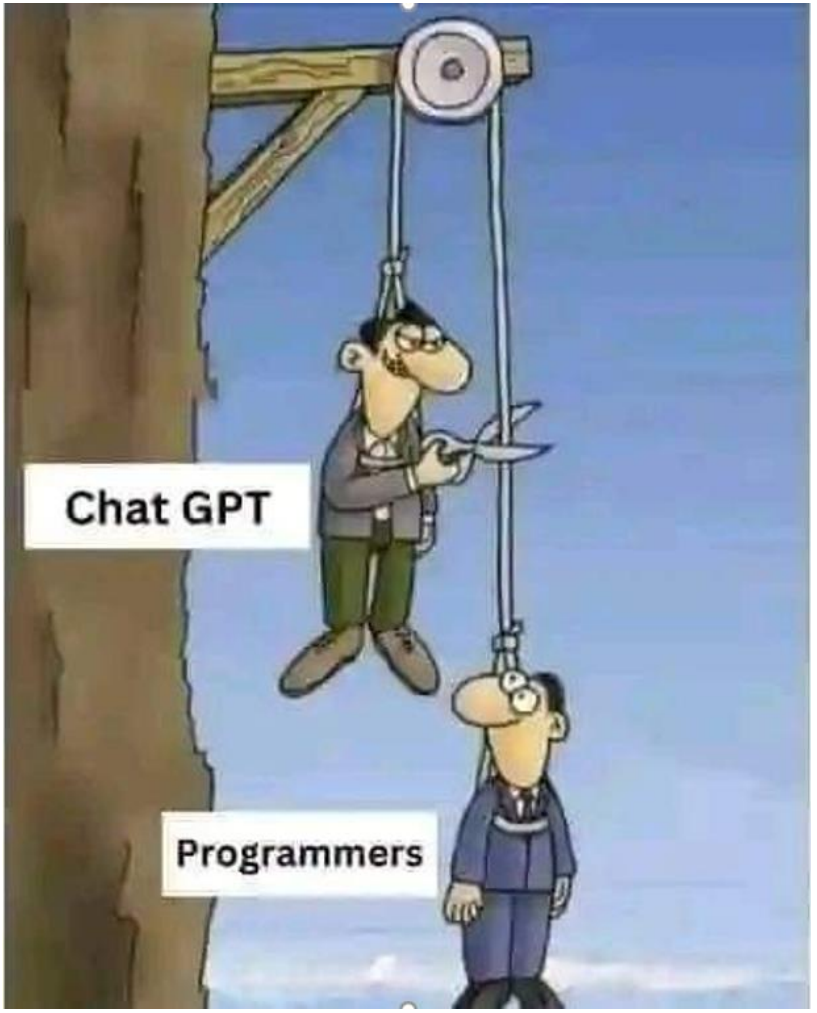
- HiddenWiki – A handy overview with lots of links to relevant dark web websites
- DuckDuckGo – The search engine of choice on the dark web
- Ahmia – A dark web search engine that allows very detailed searches
- Daniel – Another overview of very convenient dark web links
- ProPublica – A journalism platform
- SecureDrop – A safe haven for whistleblowers
- Keybase– Secure file sharing over the dark web



https://vpnoverview.com/privacy/anonymous-browsing/dark-web-websites-worth-visiting/

<참고 2> Top Programming Language on GitHub



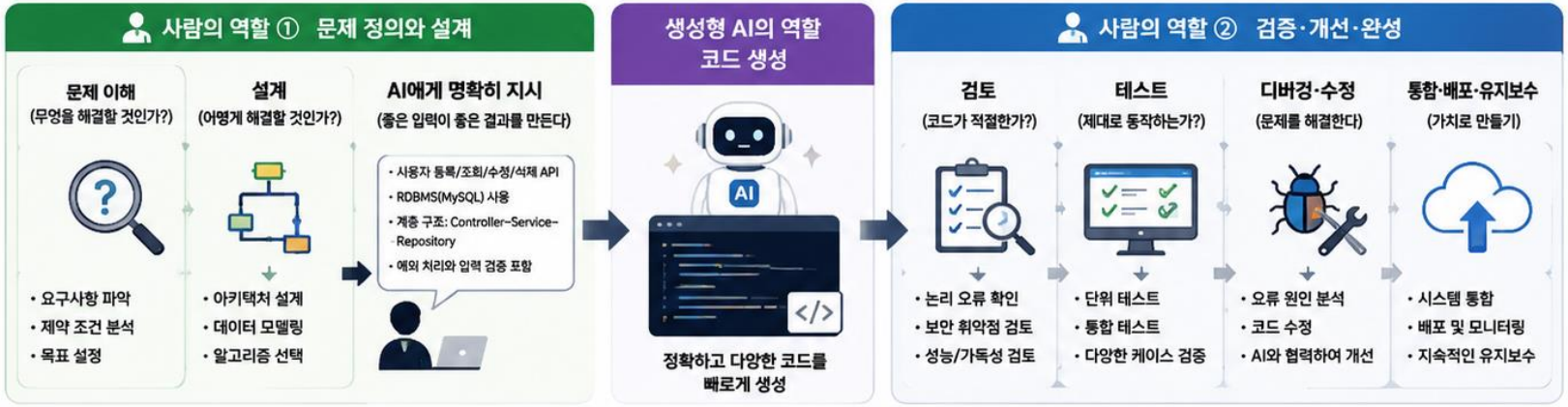


https://chatgpt.com/c/6a976e28-3864-83e9-8e3b-adbded864282

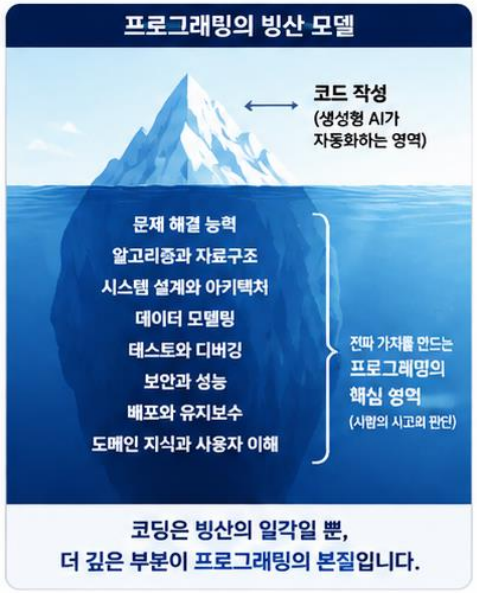
<참고 3>

생성형 AI 시대에도 프로그래밍을 배워야 하는 이유

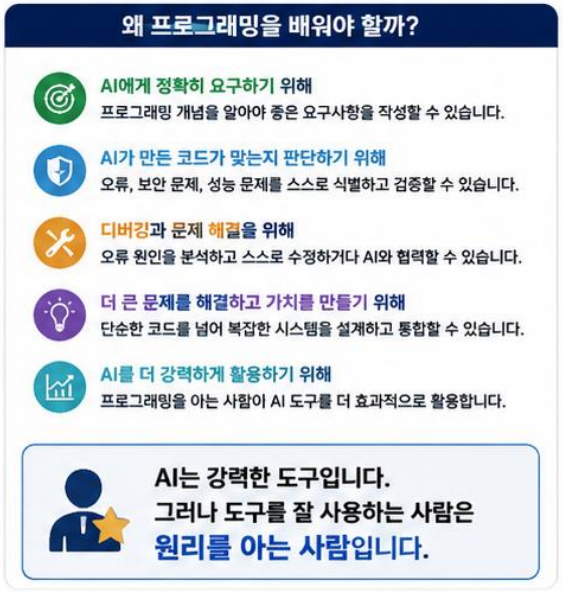
AI는 코드를 작성해 주지만, 문제를 정의하고 결과를 판단하며 가치를 만드는 것은 사람의 역할입니다.



AI는 코드를 작성할 수 있지만, 무엇을 만들지 결정하고, 코드가 올바른지 판단하며, 가치를 만들어 내는 것은 프로그래머의 역할입니다.







핵심 메시지: AI는 Coding을 자동화하지만, Programming(문제 해결과 가치 창출)을 대체하지는 않습니다.